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dataset: <https://www.kaggle.com/datasets/amar5693/screen-time-sleep-and-stress-analysis-dataset>

what data set talk about:

This dataset contains 50,000 user records analyzing smartphone usage behavior and its relationship with productivity, sleep patterns, and stress levels

```
# print top 5 rows without first row(header)
```

```
import pandas as pd
```

```
df = pd.read_csv( 'Smartphone_Usage_Productivity_Dataset_50000.csv' )  
df.head()
```

	User_ID	Age	Gender	Occupation	Device_Type	Daily_Phone_Hours
0	U1	58	Male	Professional	Android	1.3
1	U2	25	Male	Professional	Android	1.2
2	U3	19	Male	Student	iOS	5.3
3	U4	35	Female	Business Owner	iOS	5.8
4	U5	33	Male	Freelancer	Android	7.9

	Social_Media_Hours	Work_Productivity_Score	Sleep_Hours
0	6.7	6	8.8
1	1.5	5	6.4
2	5.7	5	9.0
3	2.5	2	5.7
4	1.3	4	5.7

	App_Usage_Count	Caffeine_Intake_Cups	Weekend_Screen_Time_Hours
0	42	1	8.7
1	51	3	5.1
2	14	5	6.3
3	36	6	12.8
4	37	5	9.9

```
# print the last 10 rows
```

```
df.tail(10)
```

User_ID	Age	Gender	Occupation	Device_Type
49990	24	Other	Freelancer	iOS
49991	42	Female	Business Owner	Android
49992	34	Male	Professional	iOS
49993	38	Other	Freelancer	Android
49994	34	Male	Student	Android
49995	44	Male	Business Owner	Android
49996	42	Other	Business Owner	Android
49997	27	Female	Freelancer	iOS
49998	41	Female	Business Owner	iOS
49999	46	Other	Freelancer	Android

Social_Media_Hours	Work_Productivity_Score	Sleep_Hours
49990	7.4	9
49991	1.4	6
49992	4.1	3
49993	3.1	1
49994	7.0	4
49995	5.4	5
49996	7.4	9
49997	2.5	4
49998	3.0	6
49999	2.8	3

App_Usage_Count	Caffeine_Intake_Cups
49990	1

49991	33	5
8.6		
49992	36	3
5.7		
49993	18	4
4.0		
49994	44	3
7.4		
49995	11	5
3.0		
49996	20	4
6.2		
49997	39	4
5.1		
49998	51	0
9.2		
49999	54	5
6.9		

```
# inof about dataset
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 50000 entries, 0 to 49999
```

```
Data columns (total 13 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	User_ID	50000 non-null	object
1	Age	50000 non-null	int64
2	Gender	50000 non-null	object
3	Occupation	50000 non-null	object
4	Device_Type	50000 non-null	object
5	Daily_Phone_Hours	50000 non-null	float64
6	Social_Media_Hours	50000 non-null	float64
7	Work_Productivity_Score	50000 non-null	int64
8	Sleep_Hours	50000 non-null	float64
9	Stress_Level	50000 non-null	int64
10	App_Usage_Count	50000 non-null	int64
11	Caffeine_Intake_Cups	50000 non-null	int64
12	Weekend_Screen_Time_Hours	50000 non-null	float64

```
dtypes: float64(4), int64(5), object(4)
```

```
memory usage: 5.0+ MB
```

```
# print column names
```

```
df.columns
```

```
Index(['User_ID', 'Age', 'Gender', 'Occupation', 'Device_Type',
      'Daily_Phone_Hours', 'Social_Media_Hours',
      'Work_Productivity_Score',
      'Sleep_Hours', 'Stress_Level', 'App_Usage_Count',
```

```
'Caffeine_Intake_Cups', 'Weekend_Screen_Time_Hours'],
dtype='object')
```

```
# made a subset of columns and select 5 random rows
```

```
df_1 = df[['User_ID' , 'Age' , 'Gender']].sample(n=5)
```

```
df_2 = df[['User_ID' , 'Age' , 'Gender']].sample(n=5)
```

```
df_3 = df[['User_ID' , 'Age' , 'Gender']].sample(n=5)
```

```
# print created subsets
```

```
print(f"{df_1}\n\n{df_2}\n\n{df_3}")
```

	User_ID	Age	Gender
38704	U38705	36	Male
8170	U8171	23	Other
44350	U44351	29	Male
46591	U46592	21	Female
45579	U45580	37	Male

	User_ID	Age	Gender
26571	U26572	20	Female
49199	U49200	56	Female
37504	U37505	46	Male
1908	U1909	18	Female
49520	U49521	38	Female

	User_ID	Age	Gender
35924	U35925	38	Other
47509	U47510	31	Female
15918	U15919	58	Female
8800	U8801	38	Female
17635	U17636	20	Male

```
# horizontal concatination
```

```
horizontal_concatination = pd.concat( [df_1 , df_2 , df_3] ,axis=1 )
horizontal_concatination
```

	User_ID	Age	Gender	User_ID	Age	Gender	User_ID	Age
Gender								
38704	U38705	36.0	Male	NaN	NaN	NaN	NaN	NaN
NaN								
8170	U8171	23.0	Other	NaN	NaN	NaN	NaN	NaN
NaN								
44350	U44351	29.0	Male	NaN	NaN	NaN	NaN	NaN
NaN								
46591	U46592	21.0	Female	NaN	NaN	NaN	NaN	NaN
NaN								
45579	U45580	37.0	Male	NaN	NaN	NaN	NaN	NaN
NaN								
26571	NaN	NaN	NaN	U26572	20.0	Female	NaN	NaN
NaN								
49199	NaN	NaN	NaN	U49200	56.0	Female	NaN	NaN

NaN								
37504	NaN	NaN	NaN	U37505	46.0	Male	NaN	NaN
NaN								
1908	NaN	NaN	NaN	U1909	18.0	Female	NaN	NaN
NaN								
49520	NaN	NaN	NaN	U49521	38.0	Female	NaN	NaN
NaN								
35924	NaN	NaN	NaN	NaN	NaN	NaN	U35925	38.0
Other								
47509	NaN	NaN	NaN	NaN	NaN	NaN	U47510	31.0
Female								
15918	NaN	NaN	NaN	NaN	NaN	NaN	U15919	58.0
Female								
8800	NaN	NaN	NaN	NaN	NaN	NaN	U8801	38.0
Female								
17635	NaN	NaN	NaN	NaN	NaN	NaN	U17636	20.0
Male								

vertical oncatination

```
vertical_oncatination = pd.concat( [df_1 , df_2 , df_3], axis=0 )
vertical_oncatination
```

	User_ID	Age	Gender
38704	U38705	36	Male
8170	U8171	23	Other
44350	U44351	29	Male
46591	U46592	21	Female
45579	U45580	37	Male
26571	U26572	20	Female
49199	U49200	56	Female
37504	U37505	46	Male
1908	U1909	18	Female
49520	U49521	38	Female
35924	U35925	38	Other
47509	U47510	31	Female
15918	U15919	58	Female
8800	U8801	38	Female
17635	U17636	20	Male

select the first 5 eows

```
df_1 = df[['User_ID']][:5]
df_2 = df[['Age']][:5]
df_3 = df[['Gender']][:5]
```

```
df_1 , df_2 , df_3
```

```
(  User_ID
0      U1
1      U2
2      U3
```

```

3      U4
4      U5,
   Age
0      58
1      25
2      19
3      35
4      33,
   Gender
0      Male
1      Male
2      Male
3      Female
4      Male)

```

horizontal concatenation

```

horizontal_concatination = pd.concat([df_1 , df_2 , df_3] ,axis=1)
horizontal_concatination

```

	User_ID	Age	Gender
0	U1	58	Male
1	U2	25	Male
2	U3	19	Male
3	U4	35	Female
4	U5	33	Male

statical summary about dataset

```
df.describe()
```

	Age	Daily_Phone_Hours	Social_Media_Hours \
count	50000.000000	50000.000000	50000.000000
mean	39.034960	6.509116	4.267250
std	12.414877	3.170903	2.164743
min	18.000000	1.000000	0.500000
25%	28.000000	3.800000	2.400000
50%	39.000000	6.500000	4.300000
75%	50.000000	9.200000	6.100000
max	60.000000	12.000000	8.000000

	Work_Productivity_Score	Sleep_Hours	Stress_Level
App_Usage_Count \			
count	50000.000000	50000.000000	50000.000000
50000.000000			
mean	5.503760	6.497744	5.504500
32.43898			
std	2.874806	1.449551	2.871095
16.12151			
min	1.000000	4.000000	1.000000
5.000000			
25%	3.000000	5.200000	3.000000

18.000000			
50%	5.500000	6.500000	6.000000
32.000000			
75%	8.000000	7.800000	8.000000
46.000000			
max	10.000000	9.000000	10.000000
60.000000			

	Caffeine_Intake_Cups	Weekend_Screen_Time_Hours
count	50000.000000	50000.000000
mean	3.004580	8.006844
std	1.996657	3.461300
min	0.000000	2.000000
25%	1.000000	5.000000
50%	3.000000	8.000000
75%	5.000000	11.000000
max	6.000000	14.000000

catigorical attributes

```
Device_Type = pd.DataFrame({'Device_Type':
df['Device_Type'].value_counts()})
Device_Type
```

	Device_Type
Device_Type	
Android	25080
iOS	24920

TASK 2

detect missing values:
df.isnull()

	User_ID	Age	Gender	Occupation	Device_Type
Daily_Phone_Hours \					
0	False	False	False	False	False
False					
1	False	False	False	False	False
False					
2	False	False	False	False	False
False					
3	False	False	False	False	False
False					
4	False	False	False	False	False
False					
...	...	...	...	...	...
...					
49995	False	False	False	False	False

False					
49996	False	False	False	False	False
False					
49997	False	False	False	False	False
False					
49998	False	False	False	False	False
False					
49999	False	False	False	False	False
False					

	Social_Media_Hours	Work_Productivity_Score	Sleep_Hours
Stress_Level \			
0	False	False	False
False			
1	False	False	False
False			
2	False	False	False
False			
3	False	False	False
False			
4	False	False	False
False			
...	...	...	...
...			
49995	False	False	False
False			
49996	False	False	False
False			
49997	False	False	False
False			
49998	False	False	False
False			
49999	False	False	False
False			

	App_Usage_Count	Caffeine_Intake_Cups
Weekend_Screen_Time_Hours		
0	False	False
False		
1	False	False
False		
2	False	False
False		
3	False	False
False		
4	False	False
False		
...	...	...
.		..

49995	False	False
False		
49996	False	False
False		
49997	False	False
False		
49998	False	False
False		
49999	False	False
False		

[50000 rows x 13 columns]

count missing values

df.isnull().sum()

User_ID	0
Age	0
Gender	0
Occupation	0
Device_Type	0
Daily_Phone_Hours	0
Social_Media_Hours	0
Work_Productivity_Score	0
Sleep_Hours	0
Stress_Level	0
App_Usage_Count	0
Caffeine_Intake_Cups	0
Weekend_Screen_Time_Hours	0

dtype: int64

df.select_dtypes(include= ['int64' , 'float64']).columns

Index(['Age', 'Daily_Phone_Hours', 'Social_Media_Hours',
'Work_Productivity_Score', 'Sleep_Hours', 'Stress_Level',
'App_Usage_Count', 'Caffeine_Intake_Cups',
'Weekend_Screen_Time_Hours'],
dtype='object')

handle outliers

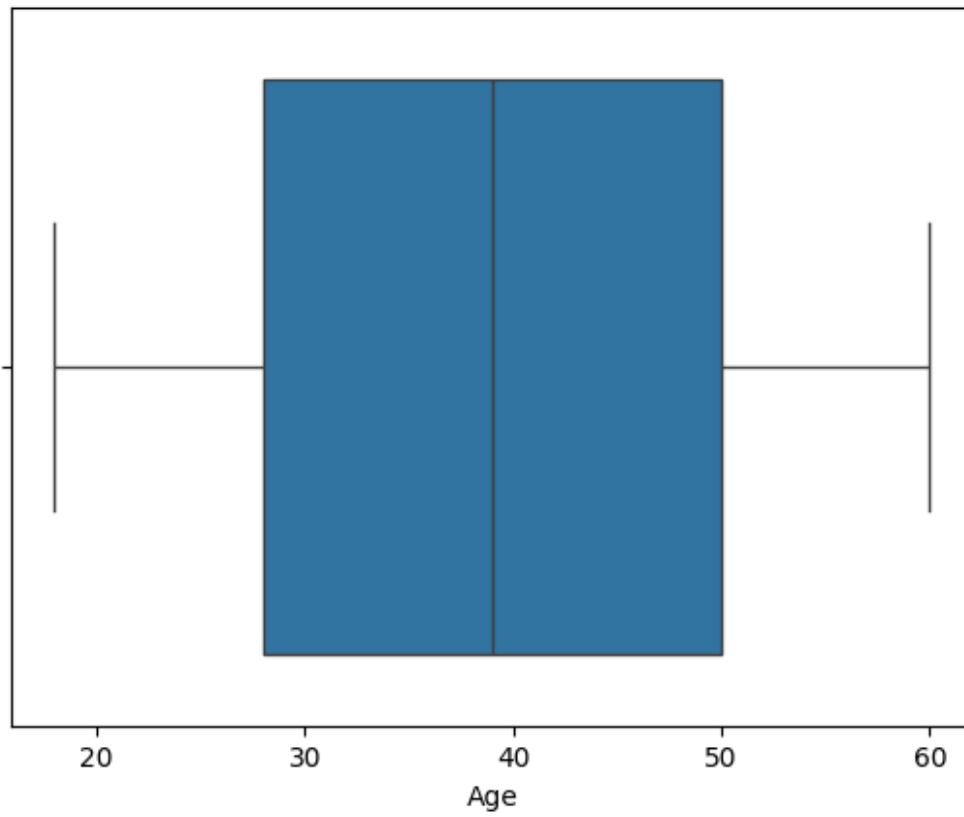
import seaborn as sns

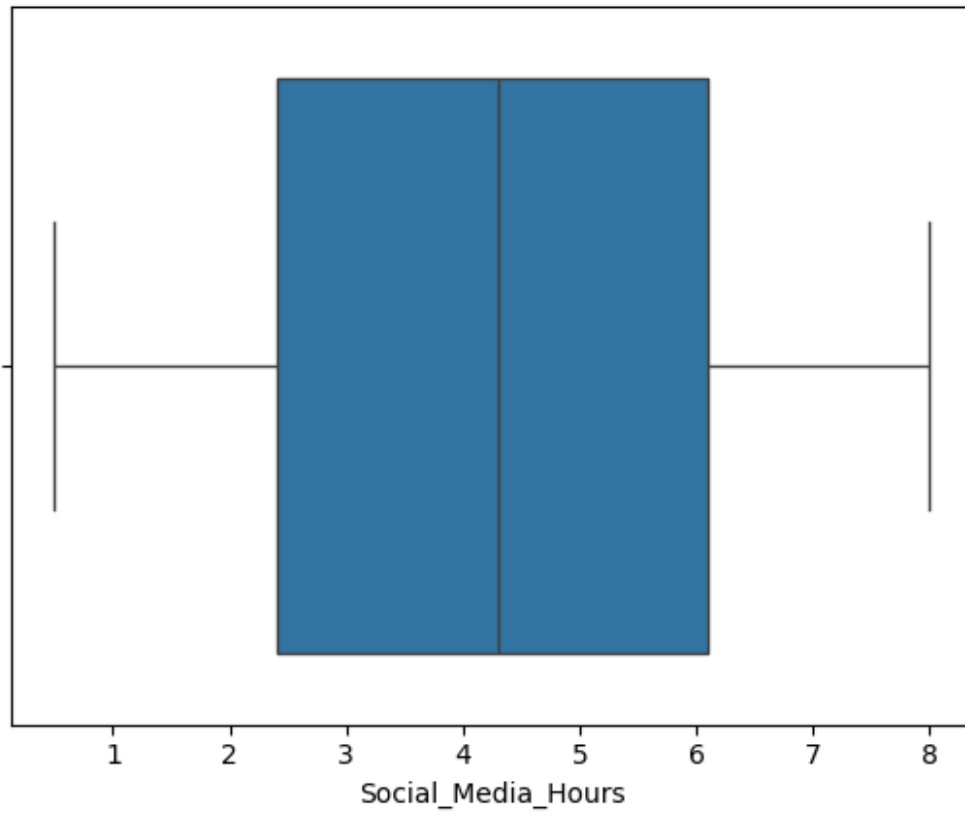
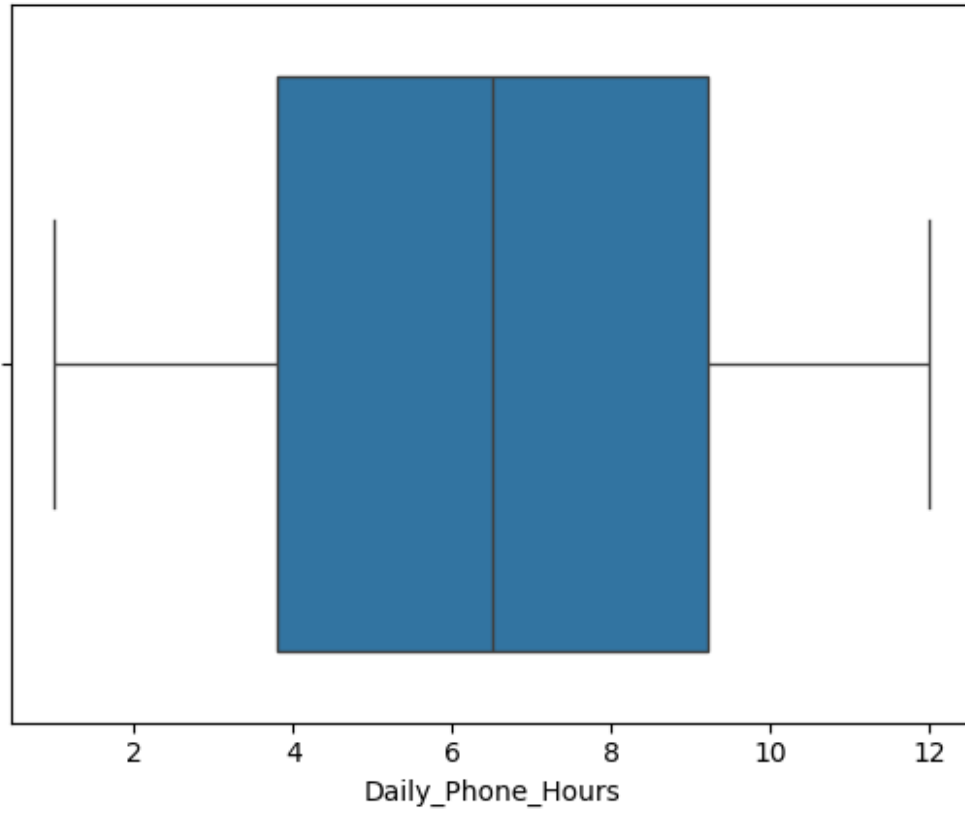
import matplotlib.pyplot as plt

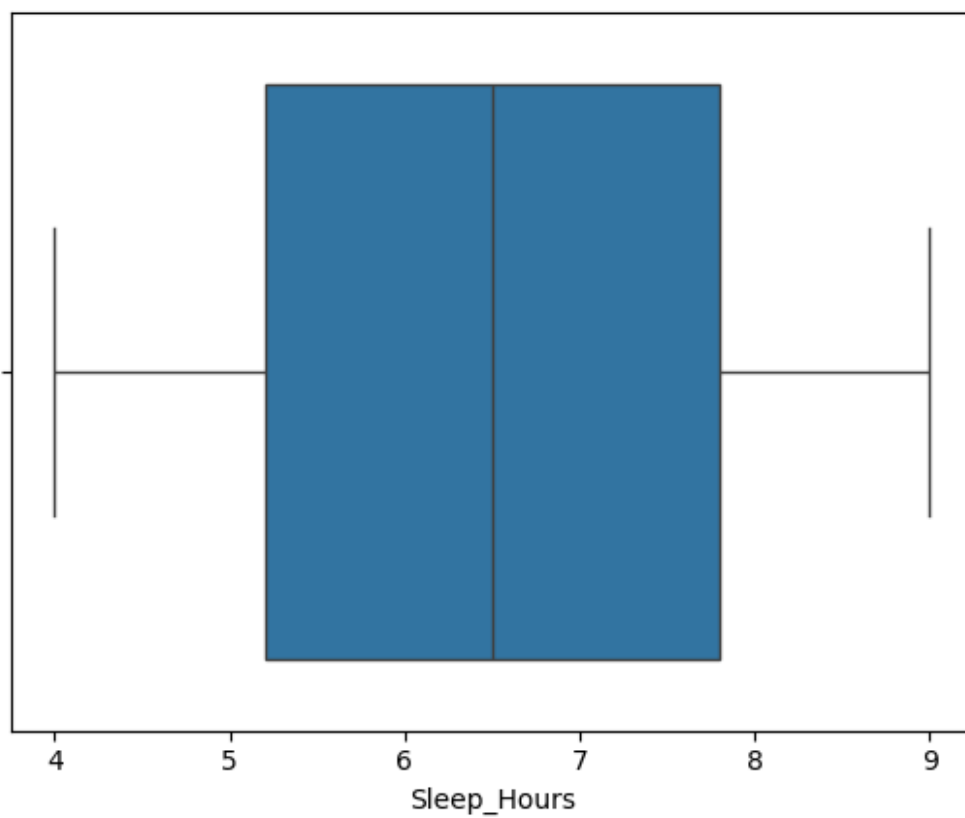
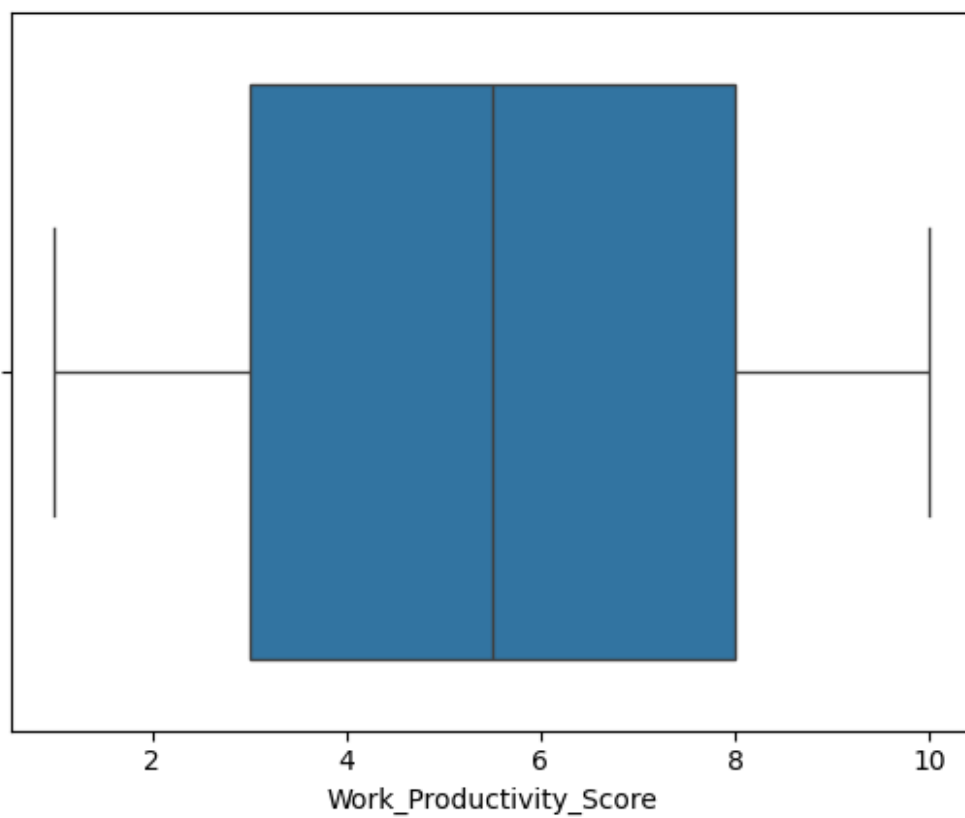
num_cols = ['Age', 'Daily_Phone_Hours', 'Social_Media_Hours',
'Work_Productivity_Score', 'Sleep_Hours', 'Stress_Level',
'App_Usage_Count', 'Caffeine_Intake_Cups',
'Weekend_Screen_Time_Hours']

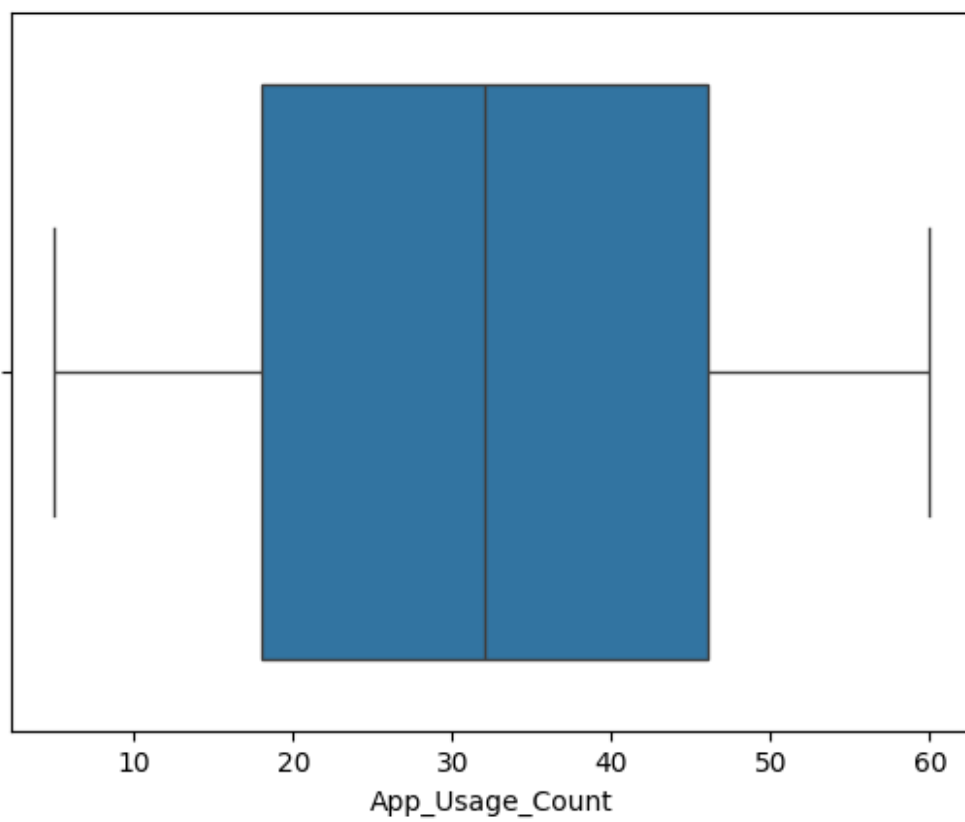
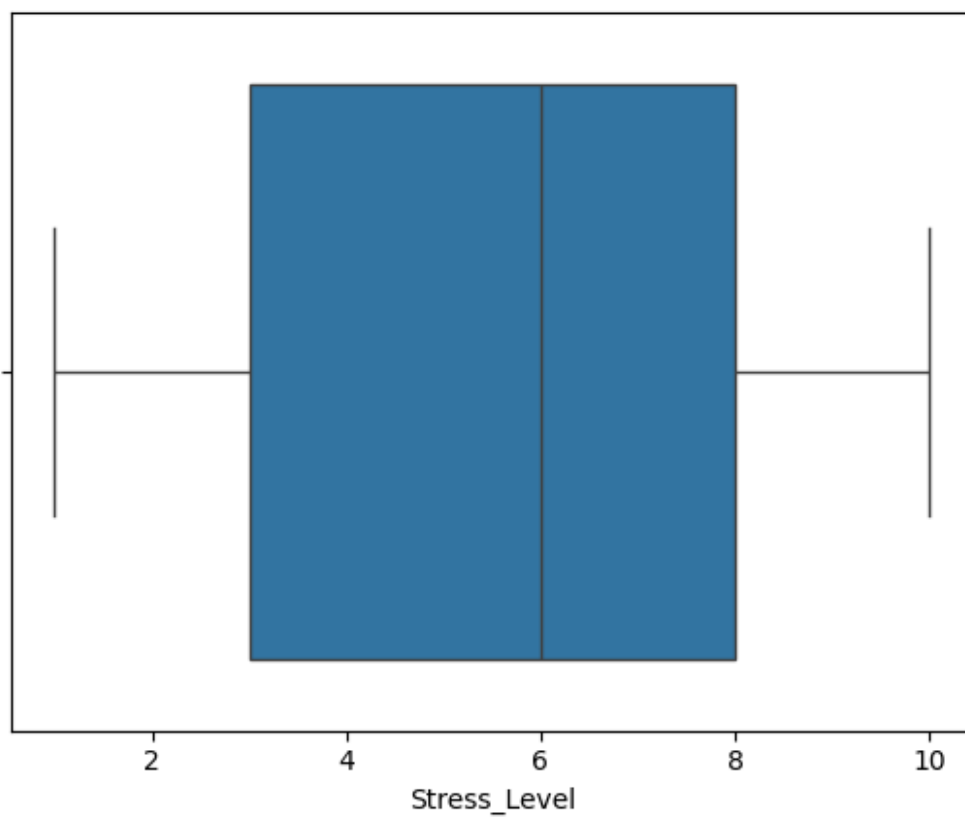
for col in num_cols:

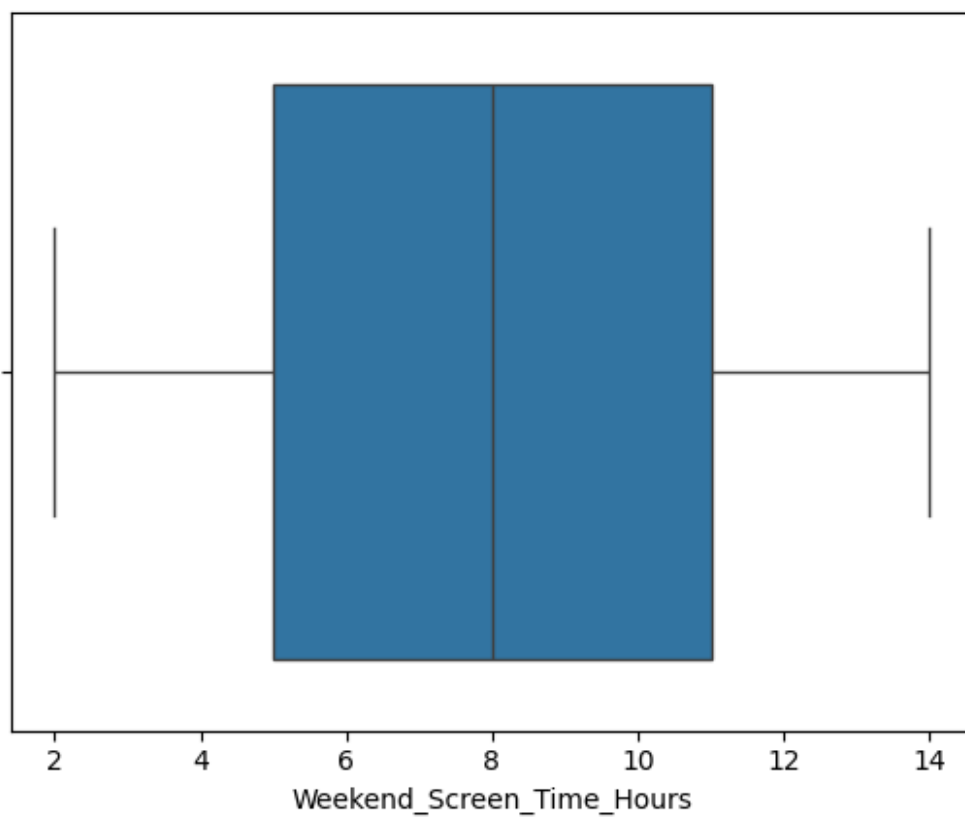
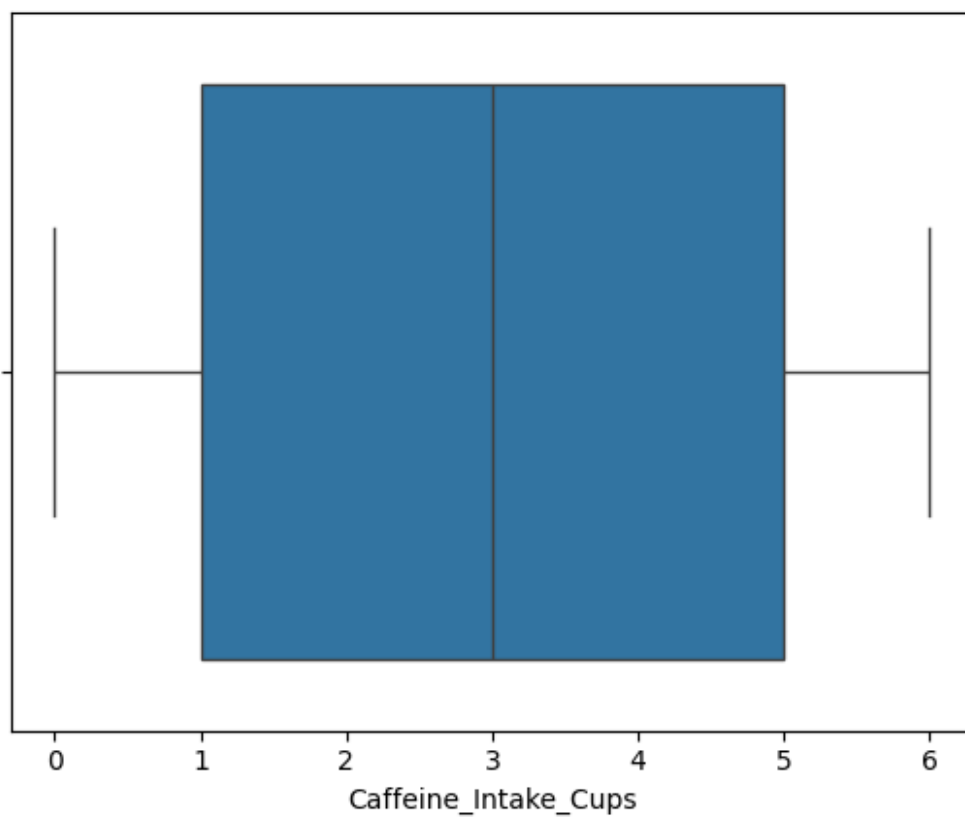
```
sns.boxplot(x = df[col])  
plt.show()
```











```
# 1 Data Discretization (Converting Continuous Values to Categories)
num_cols = ['Age', 'Daily_Phone_Hours', 'Social_Media_Hours',
            'Work_Productivity_Score', 'Sleep_Hours', 'Stress_Level',
            'App_Usage_Count', 'Caffeine_Intake_Cups',
            'Weekend_Screen_Time_Hours']
```

```
import pandas as pd
```

```
num_cols = ['Age', 'Daily_Phone_Hours', 'Social_Media_Hours',
            'Work_Productivity_Score', 'Sleep_Hours', 'Stress_Level',
            'App_Usage_Count', 'Caffeine_Intake_Cups',
            'Weekend_Screen_Time_Hours']
```

```
n_bins = 3
```

```
labels = ['Low', 'Medium', 'High']
```

```
for col in num_cols:
    df[col + '_binned'] = pd.qcut(df[col], q=n_bins, labels=labels)
```

```
df.head()
```

	User_ID	Age	Gender	Occupation	Device_Type	Daily_Phone_Hours
0	U1	58	Male	Professional	Android	1.3
1	U2	25	Male	Professional	Android	1.2
2	U3	19	Male	Student	iOS	5.3
3	U4	35	Female	Business Owner	iOS	5.8
4	U5	33	Male	Freelancer	Android	7.9

	Social_Media_Hours	Work_Productivity_Score	Sleep_Hours
0	6.7	6	8.8
4			
1	1.5	5	6.4
1			
2	5.7	5	9.0
4			
3	2.5	2	5.7
3			
4	1.3	4	5.7
3			

	Weekend_Screen_Time_Hours	Age_binned
0	8.7	High

Low		
1 ...	5.1	Low
Low		
2 ...	6.3	Low
Medium		
3 ...	12.8	Medium
Medium		
4 ...	9.9	Medium
Medium		

Social_Media_Hours_binned Work_Productivity_Score_binned		
Sleep_Hours_binned \		
0	High	Medium
High		
1	Low	Medium
Medium		
2	High	Medium
High		
3	Low	Low
Low		
4	Low	Low
Low		

Stress_Level_binned App_Usage_Count_binned		
Caffeine_Intake_Cups_binned \		
0	Low	Medium
Low		
1	Low	High
Medium		
2	Low	Low
High		
3	Low	Medium
High		
4	Low	Medium
High		

Weekend_Screen_Time_Hours_binned	
0	Medium
1	Low
2	Medium
3	High
4	Medium

[5 rows x 22 columns]

```
# Normalization (Min-Max Scaling)
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
```



```
df_scaled = df.copy()
df_scaled[num_cols] = scaler.fit_transform(df[num_cols])

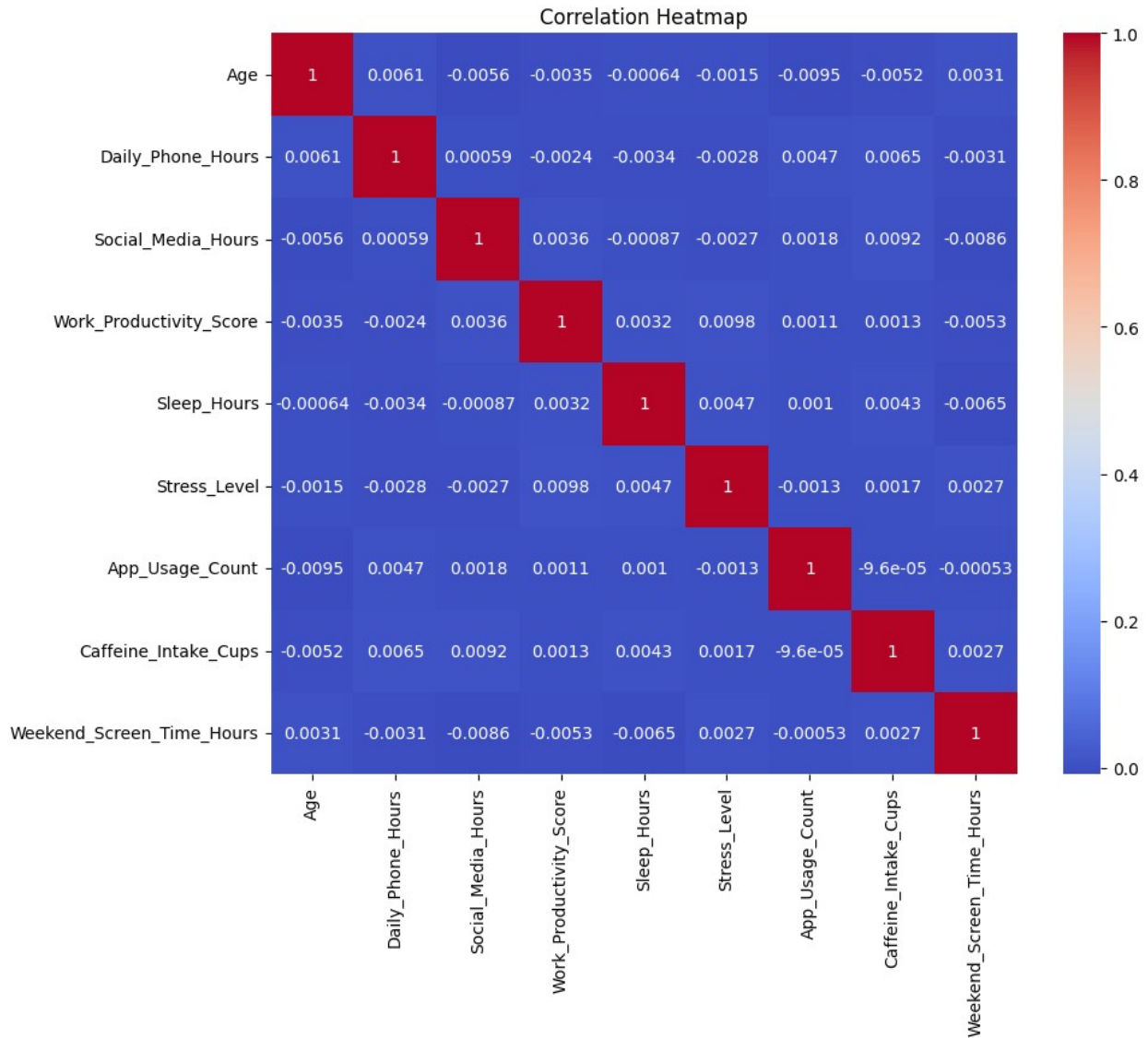
df_scaled[num_cols].min()
df_scaled[num_cols].max()

Age                                1.0
Daily_Phone_Hours                 1.0
Social_Media_Hours                1.0
Work_Productivity_Score           1.0
Sleep_Hours                       1.0
Stress_Level                      1.0
App_Usage_Count                   1.0
Caffeine_Intake_Cups              1.0
Weekend_Screen_Time_Hours         1.0
dtype: float64

# Visualize relationships using appropriate plots
import seaborn as sns
import matplotlib.pyplot as plt

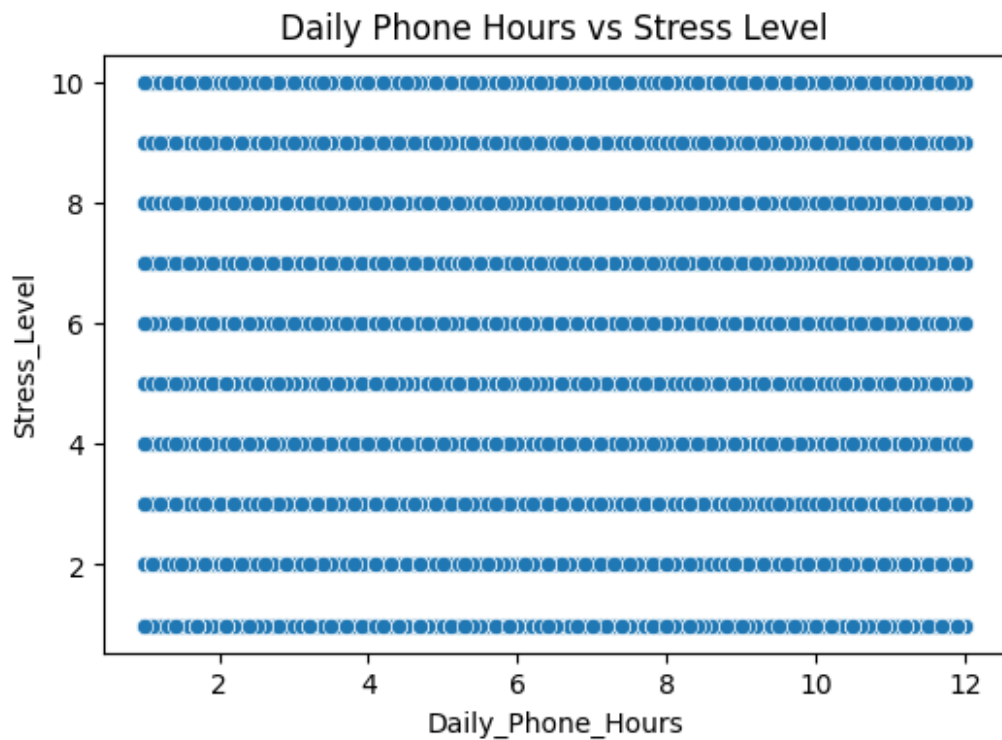
plt.figure(figsize=(10,8))
corr = df[num_cols].corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
sns.scatterplot(data=df, x='Daily_Phone_Hours', y='Stress_Level')
plt.title("Daily Phone Hours vs Stress Level")
plt.show()
```



```
df[['Daily_Phone_Hours', 'Stress_Level']].corr()
```

	Daily_Phone_Hours	Stress_Level
Daily_Phone_Hours	1.000000	-0.002762
Stress_Level	-0.002762	1.000000